

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

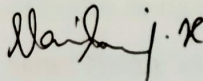
Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I ~~VAIBHAVA LAKSHMI.K~~ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

CO₂ - AT1

Analytical Problem Solving

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COURSE: COMPUTER NETWORKS

COURSE CODE: CSA0714

1) Given:

Network: 192.168.20.0/24

The Network is divided into:

- Lab A $\rightarrow$ /26
- Lab B $\rightarrow$ /27
- Lab C $\rightarrow$ /28

Subnet Allocation:

$\rightarrow$ Lab A (/26)

- Subnet Mask: 255.255.255.192

- Number of IP Addresses:

$$2^{(32-26)} = 64$$

$$64 - 2 = \underline{62}$$

Network Address: 192.168.20.0

- First Host: 192.168.20.1
- Last Host: 192.168.20.62

$\Rightarrow$ Broadcast Host: 192.168.20.63

→ Lab B (127)

- Subnet Mask: 255.255.255.224
- Number of IP Addresses:

$$2^{(32-27)} = 32$$

$$32 - 2 = \underline{30}$$

Network Address : 192.168.20.64

- First Host : 192.168.20.65
- Last Host : 192.168.20.94

⇒ Broadcast Address : 192.168.20.95

→ Lab C (128)

- Subnet Mask: 255.255.255.240
- Number of IP Addresses:

$$2^{(32-28)} = 16$$

$$16 - 2 = \underline{14}$$

Network Address : 192.168.20.96

- First Host : 192.168.20.97
- Last Host : 192.168.20.110

⇒ Broadcast Address = 192.168.20.111

Conclusion:

* The university network $192.168.20.0/24$ is successfully divided into 3 subnets for Lab A, B & C.

* Lab A receives $1/26$ subnet with 62 host address.

* Lab B receives $1/27$ subnet with 30 host address

* Lab C receives $1/28$ subnet with 14 host address.

3)

Given:

• IP Address Block: 172.16.0.0/16

• Departments & Subnet Masks:

→ HR = 125

→ Sales = 126

→ IT = 127

→ Admin = 128

Subnet Allocation:

→ HR Department (125)

• Subnet Mask: 255.255.255.128

• Total IP addresses:

$$2^{(32-25)} = 128$$

$$128 - 2 = \underline{126}$$

Network Address : 172.16.0.0

• First Host : 172.16.0.1

• Last Host : 172.16.0.126

⇒ Broadcast Address : 172.16.0.127

→ Sales Department (126)

• Subnet Mask : 255.255.255.192

• Total IP addresses:

$$2^{(32-26)} = 64$$

$$64 - 2 = \underline{62}$$

Network Address : 172.16.0.128

• First Host : 172.16.0.129

• Last Host : 172.16.0.190

⇒ Broadcast Address : 172.16.0.191

→ IT Department (127):

• Subnet mask : 255.255.255.224

• Total IP addresses:

$$2^{(32-27)} = 32$$

$$32 - 2 = \underline{30}$$

Network Address : 172.16.0.192

• First Host : 172.16.0.193

• Last Host : 172.16.0.222

⇒ Broadcast Address : 172.16.0.223

→ Admin Department (128)

• Subnet Mask : 255.255.255.240

• Total IP addresses:

$$2^{(32-28)} = 16$$

$$16 - 2 = \underline{14}$$

Network Address : 172.16.0.224

• First Host : 172.16.0.225

• Last Host : 172.16.0.238.

⇒ Broadcast Address : 172.16.0.239.